MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, LOW-POWER SCHOTTKY, TTL, COUNTERS, MONOLITHIC SILICON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, low-power Schottky, TTL, binary and decade counter microcircuits. Three product assurance classes and a choice of case outlines and lead finishes are provided for each type and are reflected in the part number.

1.2 Part number. The part number shall be in accordance with MIL-M-38510, and as specified herein.

1.2.1 Device type. The device type shall be as follows:

Device type	Circuit
01	Dual decade counter with A and B inputs
02	Dual 4-bit binary counter
03	Dual decade counter with clear and set-to-nine

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outline. The case outline shall be designated as follows:

Letter	Case outline (see MIL-M-38510, appendix C)
A	F-1 (14-lead, 1/4" x 1/4") flat package
B	F-3 (14-lead, 3/16" x 1/4") flat package
C	D-1 (14-lead, 1/4" x 3/4") dual-in-line package
D	F-2 (14-lead, 1/4" x 3/8") flat package
E	D-2 (16-lead, 1/4" x 7/8") dual-in-line package
F	F-5 (16-lead, 1/4" x 3/8") flat package

1.3 Absolute maximum ratings.

Supply voltage range	-0.5 V to +7.0 V
Input voltage range	-1.5 V at -18 mA to +5.5 V
Storage temperature range	-65°C to +150°C
Maximum power dissipation (P_D) 1/	143 mW
Lead temperature (soldering, 10 seconds)	300°C
Thermal resistance, junction to case (θ_{JC}):	
Cases A, B, D, and F	0.09°C/mW
Cases C and E	0.08°C/mW
Junction temperature (T_J)	175°C

1/ Must withstand the added P_D due to short circuit test (e.g., I_{OS}).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center (RBE-2), Griffiss AFB, NY 13441, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.4 Recommended operating conditions.

Supply voltage	4.5 V minimum to 5.5 V maxim
Minimum high-level input voltage	2.0 V
Maximum low-level input voltage	0.7 V
Case operating temperature range (T_C)	-55°C to +125°C
Normalized fanout (each output)	10 maximum
Width of input count pulse, $t_{p(IN)}$:	
Input A, clock	20 ns minimum
Input B	25 ns minimum
Width of clear pulse	20 ns minimum
Input clock frequency (F_{MAX}):	
Input A (Types 01, 02, 03)	0 to 25 MHz
Input B (Type 01)	0 to 12.5 MHz
Setup time (Types 01 and 02)	
Clear inactive static setup time (time for clear ↓ to A or B input ↓)	85 ns minimum

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-M-38510, and as specfied herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510 and herein.

3.2.1 Logic diagrams and terminal connections. The logic diagrams and terminal connections shall be as specified on figures 1 and 2, respectively.

3.2.2 Truth tables. The truth tables shall be as specified on figure 3.

3.2.3 Schematic circuits. Schematic circuits shall be submitted to the preparing activity prior to inclusion of a manufacturer's device in this specification and shall be submitted to the qualifying activity as a prerequisite for qualification. All manufacturers' schematics shall be maintained and available upon request.

3.2.4 Case outlines. The case outlines shall be as specified in 1.2.3.

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TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions	Device type	Limits		Unit
				Min	Max	
High-level output voltage	V_{OH}	V_{CC} = 4.5 V, V_{IL} = 0.7 V, I_{OH} = -400 µA, V_{IH} = 2.0 V	All	2.5		V
Low-level output voltage	V_{OL}	V_{CC} = 4.5 V, V_{IH} = 2.0 V, I_{OL} = 4 mA, V_{IL} = 0.7 V	All		0.4	V
Input clamp voltage	V_{IC}	V_{CC} = 4.5 V, I_{IN} = -18 mA, T_C = 25°C	All		-1.5	V
High-level input current at clear or set to 9 inputs	I_{IH1}	V_{CC} = 5.5 V, V_{IN} = 2.7 V	All		20	µA
High-level input current at clear or set to 9 inputs	I_{IH4}	V_{CC} = 5.5 V, V_{IN} = 5.5 V	All		100	µA
High-level input current at input A or clock	I_{IH2}	V_{CC} = 5.5 V, V_{IN} = 2.7 V	All		100	µA
High-level input current at input A or clock	I_{IH5}	V_{CC} = 5.5 V, V_{IN} = 5.5 V	01,03		200	µA
			02		400	µA
High-level input current at input B	I_{IH3}	V_{CC} = 5.5 V, V_{IN} = 2.7 V	01		200	µA
High-level input current at input B	I_{IH6}	V_{CC} = 5.5 V, V_{IN} = 5.5 V	01		400	µA
Low-level input current at clear or set to 9 inputs	I_{IL1}	V_{CC} = 5.5 V, V_{IN} = 0.4 V	01,02	-120	-400	µA
			03	-135	-400	µA
Low-level input current at input A	I_{IL2}	V_{CC} = 5.5 V, V_{IN} = 0.4 V	All	-.35	-2.4	mA
Low-level input current at input B	I_{IL3}	V_{CC} = 5.5 V, V_{IN} = 0.4 V	01	-.60	-3.2	mA
Short-circuit supply current	I_{OS}	V_{CC} = 5.5 V 1/	All	-15	-100	mA
Supply current	I_{CC}	V_{CC} = 5.5 V, V_{IN} = 0 V	All		26	mA

1/ Not more than one output should be shorted at a time.

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions		Device type	Limits Min	Limits Max	Un.
Maximum input clock frequency, A or CLK	F_{MAX1}	V_{CC} = 5.0 V, C_L = 50 pF, R_L = 2 kΩ		All	25		MHz
Maximum input clock frequency, B	F_{MAX2}	V_C = 5.0 V, C_L = 50 pF, R_L = 2 kΩ		01	12.5		MHz
Propagation delay time, low to high	t_{PLH1}	V_{CC} = 5.0 V, C_L = 50 pF, R_L = 2 kΩ	A>Q_C	01	2	84	ns
			A>Q_A	02	2	33	ns
			CLK>Q_A	03	2	33	ns
Propagation delay time, high to low	t_{PHL1}	V_{CC} = 5.0 V, C_L = 50 pF, R_L = 2 kΩ	A>Q_C	01	2	84	ns
			A>Q_A	02	2	33	ns
			CLK>G	03	2	33	ns
Propagation delay time, low to high	t_{PLH2}	V_{CC} = 5.0 V, C_L = 50 pF, R_L = 2 kΩ	B>Q_D	01	2	34	ns
			A>Q_D	02	2	84	ns
			CLK>Q_A	03	2	77	ns
Propagation delay time, high to low	t_{PHL2}	V_{CC} = 5.0 V, C_L = 50 pF, R_L = 2 kΩ	B>Q_D	01	2	34	
			A>Q_D	02	2	84	ns
			CLK>Q_C	03	2	77	ns
Propagation delay time, high to low, CLR to Q	t_{PHL3}	V_{CC} = 5.0 V, C_L = 50 pF, R_L = 2 kΩ		All	2	56	ns
Propagation delay time, low to high, set to 9 to Q_A	t_{PLH4}	V_{CC} = 5.0 V, C_L = 50 pF, R_L = 2 kΩ		03	2	57	ns
Propagation delay time, high to low, set to 9 to Q_B	t_{PHL4}	V_{CC} = 5.0 V, C_L = 50 pF, R_L = 2 kΩ		03	2	53	ns

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3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-38510 (see 6.5).

3.4 Electrical performance characteristics. The electrical performance characteristics are as specified in table I, and apply over the full recommended case operating temperature range, unless otherwise specified.

3.5 Electrical test requirements. The electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III. (Subgroups 7 and 8 testing requires only a summary of attributes data.)

3.6 Marking. Marking shall be in accordance with MIL-M-38510. At the option of the manufacturer, marking of the country of origin may be omitted from the body of the microcircuit but shall be retained on the initial container.

3.7 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 12 (see MIL-M-38510, appendix E).

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007, as applicable, of MIL-STD-883, except as modified herein.

TABLE II. Electrical test requirements.

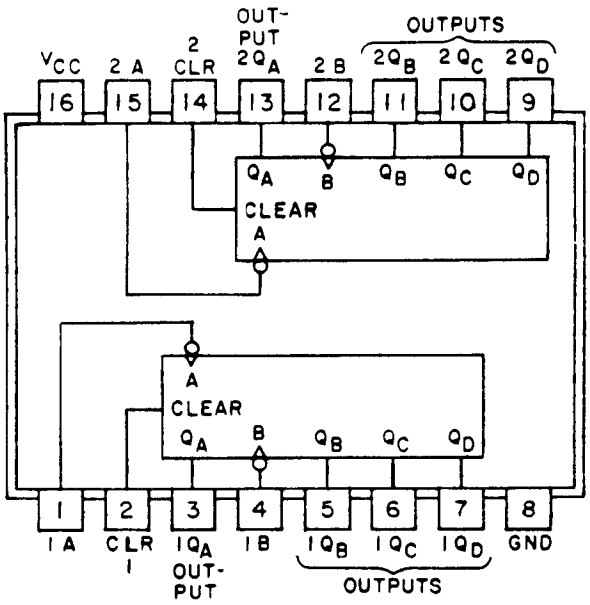
MIL-STD-883 test requirements	Subgroups (see table III)		
	Class S devices	Class B devices	Class C devices
Interim electrical parameters (pre burn-in) (method 5004)	1	1	None
Final electrical test parameters (method 5004)	1*,2,3,7, 9,10,11	1*,2,3, 7,9	1,7
Group A electrical tests (method 5005)	1,2,3,7, 8,9,10,11	1,2,3,7, 9,10,11	1,2,3, 7,9
Group B electrical tests (method 5005, subgroup 5)	1,2,3,7, 8,9,10,11	---	---
Group C end-point electrical parameters (method 5005)	---	1,2,3	1
Additional electrical tests for group C inspection	---	None	10,11
Group D end-point electrical parameters (method 5005)	1,2,3	1,2,3	1

* PDA applies to subgroup 1 (see 4.2c.).

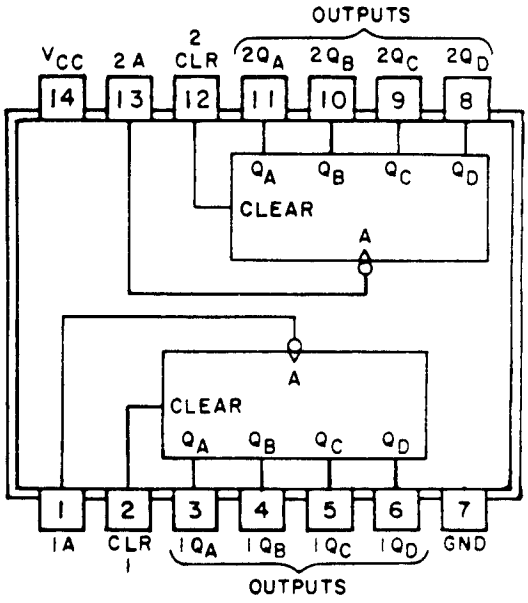
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Device type 01
Cases E and F



Device type 02
Cases A, B, C, and D



Device type 03
Cases E and F

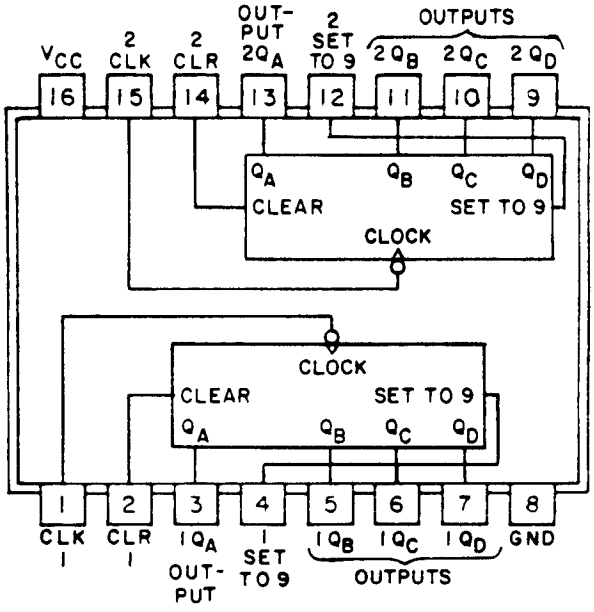


FIGURE 1. Terminal connections.

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Device type 01

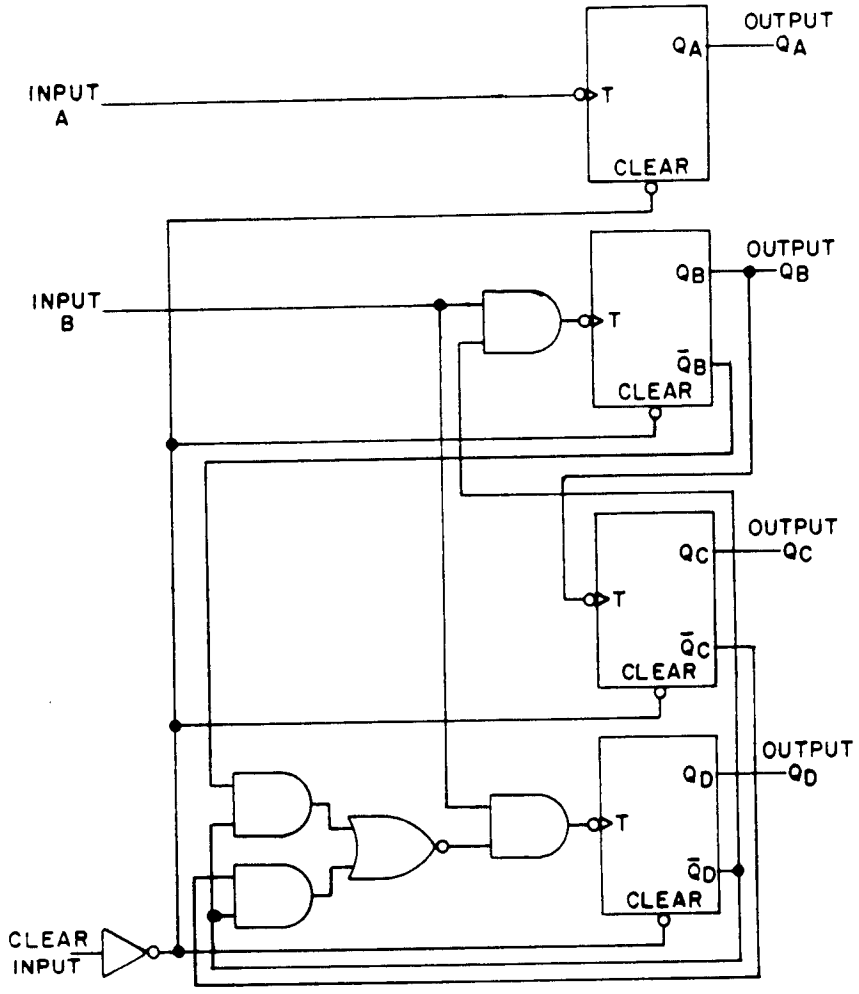


FIGURE 2. Logic diagrams (each counter).

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Device type 02

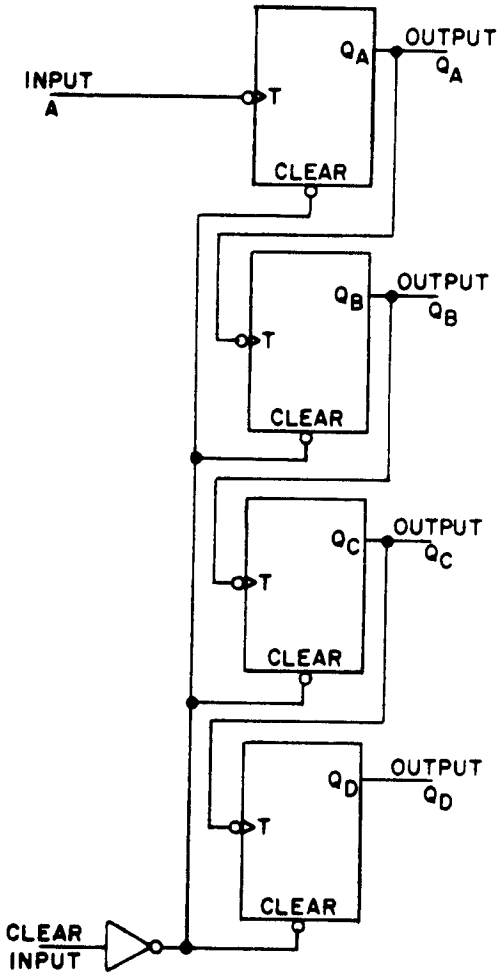


FIGURE 2. Logic diagrams (each counter) - Continued.

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Device type 03

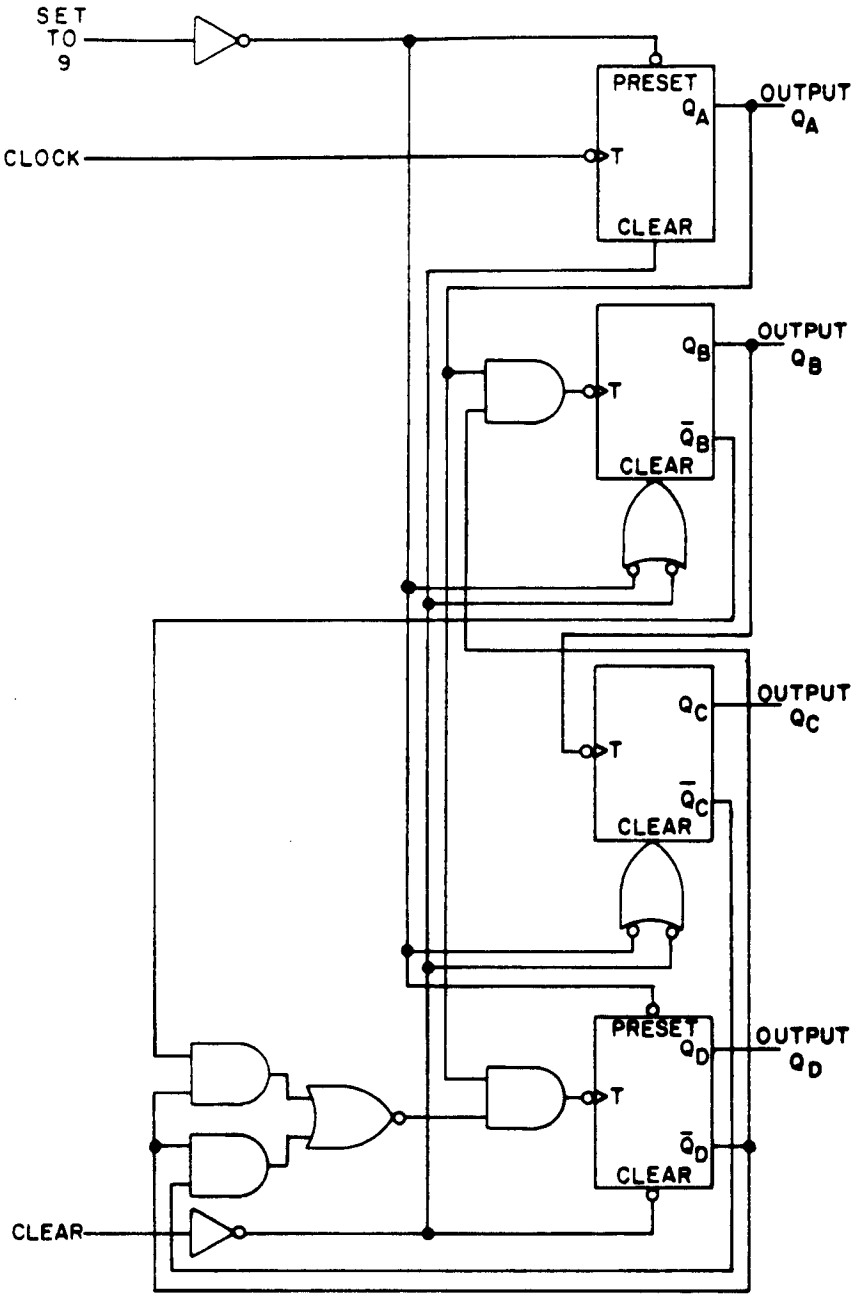


FIGURE 2. Logic diagrams (each counter) - Continued.

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Device type 01

EACH COUNTER

BCD COUNT SEQUENCE

(See Note A)

COUNT	OUTPUT			
	Q_D	Q_C	Q_B	Q_A
0	L	L	L	L
1	L	L	L	H
2	L	L	H	L
3	L	L	H	H
4	L	H	L	L
5	L	H	L	H
6	L	H	H	L
7	L	H	H	H
8	H	L	L	L
9	H	L	L	H

BI-QUINARY (5-2)

(See Note B)

COUNT	OUTPUT			
	Q_A	Q_D	Q_C	Q_B
0	L	L	L	L
1	L	L	L	H
2	L	L	H	L
3	L	L	H	H
4	L	H	L	L
5	H	L	L	L
6	H	L	L	H
7	H	L	H	L
8	H	L	H	H
9	H	H	L	L

NOTES:
A. Output Q_A is connected to input B for BCD count.
B. Output Q_D is connected to input A for bi-quinary count.

Device type 02

EACH COUNTER

COUNT SEQUENCE

COUNT	OUTPUT			
	Q_D	Q_C	Q_B	Q_A
0	L	L	L	L
1	L	L	L	H
2	L	L	H	L
3	L	L	H	H
4	L	H	L	L
5	L	H	L	H
6	L	H	H	L
7	L	H	H	H
8	H	L	L	L
9	H	L	L	H
10	H	L	H	L
11	H	L	H	H
12	H	H	L	L
13	H	H	L	H
14	H	H	H	L
15	H	H	H	H

Device type 03

BCD COUNT SEQUENCE

(EACH COUNTER)

COUNT	OUTPUT			
	Q_D	Q_C	Q_B	Q_A
0	L	L	L	L
1	L	L	L	H
2	L	L	H	L
3	L	L	H	H
4	L	H	L	L
5	L	H	L	H
6	L	H	H	L
7	L	H	H	H
8	H	L	L	L
9	H	L	L	H

CLEAR/SET TO 9

FUNCTION TABLE

(EACH COUNTER)

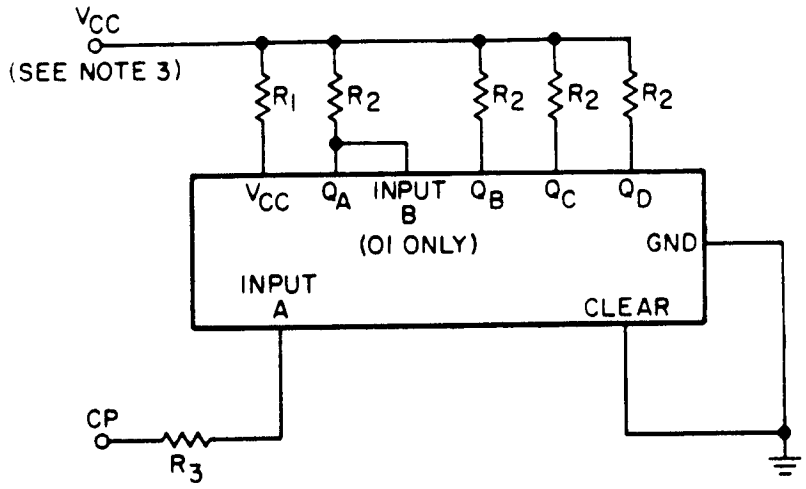
INPUTS		OUTPUTS			
CLEAR	SET TO 9	Q_A	Q_B	Q_C	Q_D
H	L	L	L	L	L
L	H	H	L	L	H
L	L	COUNT			

H = high level, L = low level

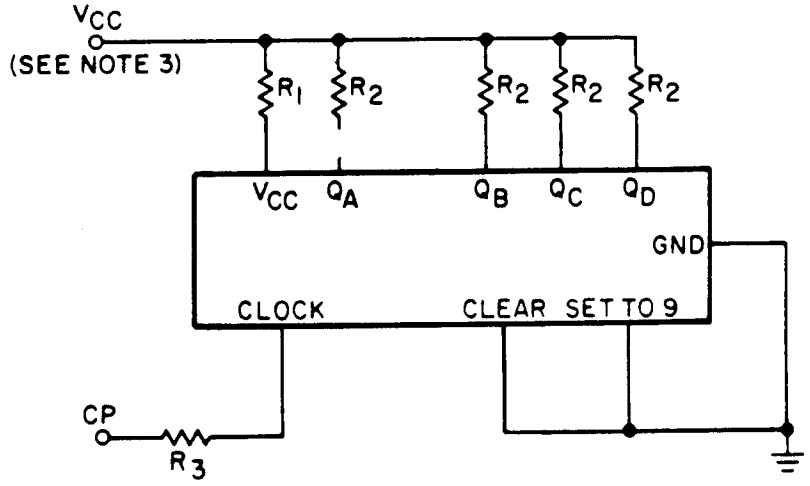
FIGURE 3. Truth tables.

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Device types 01 and 02



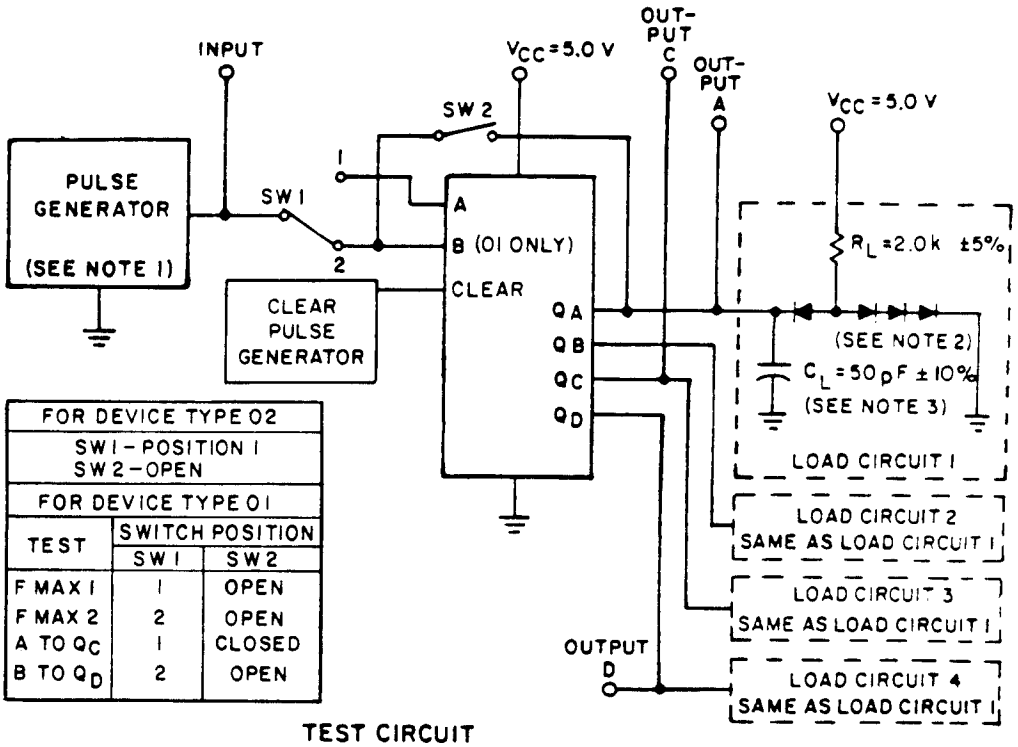
Device type 03



NOTES:
1. CP = 100 kHz ±50% square wave; duty cycle = 50 ±15%; V_{IL} = -0.5 V minimum to +0.7 V maximum; V_{IH} = 2.0 V minimum to 5.5 V maximum.
2. R_1 = 20Ω (max), R_2 = 1.0 kΩ ±10%, R_3 = 27Ω ±10%.
3. V_{CC} shall be high enough to insure that 5.0 V minimum is present at V_{CC} device terminal.

FIGURE 4. Burn-in and life test circuit.

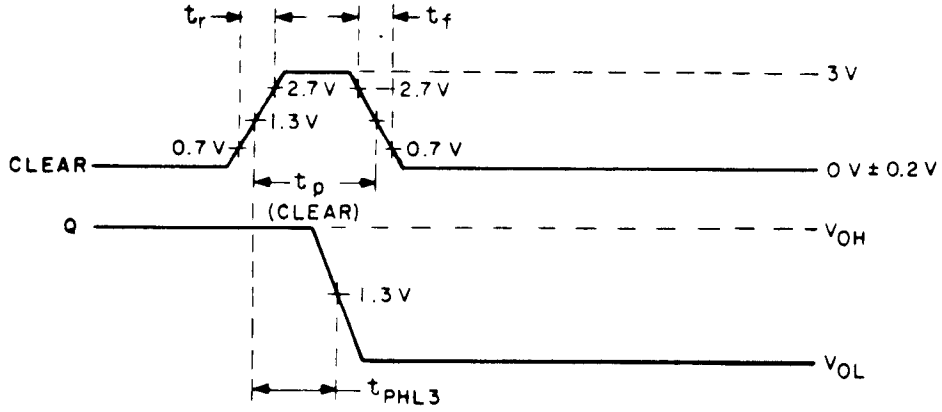
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FOR DEVICE TYPE 02		
SW 1 – POSITION 1 SW 2 – OPEN		
FOR DEVICE TYPE 01		
TEST	SWITCH POSITION	
	SW 1	SW 2
F MAX 1	1	OPEN
F MAX 2	2	OPEN
A TO Q_C	1	CLOSED
B TO Q_D	2	OPEN

TEST CIRCUIT

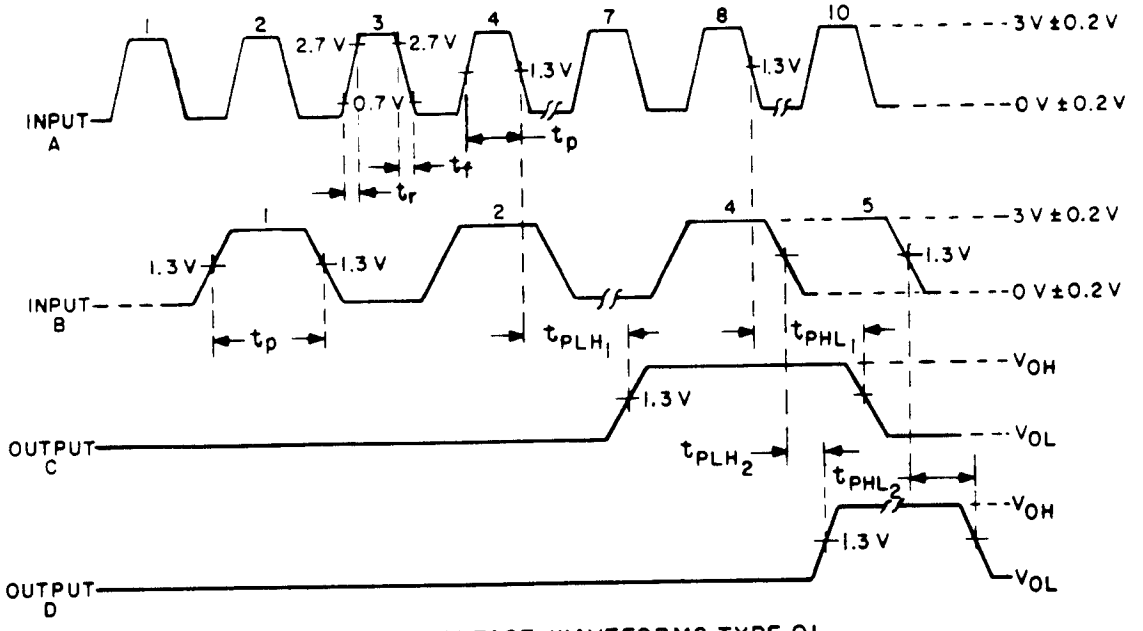
NOTES:
1. The pulse generator has the following characteristics: V_{gen} = 3 V, $t_r \leq 15$ ns, $t_f \leq 6$ ns, t_p = .5 µs, PRR ≤ 1 MHz, $Z_{out} \approx 50$, $t_{p(clear)} \geq 20$ ns.
2. All diodes are 1N3064 or equivalent.
3. C_L includes probe and jig capacitance.
4. Voltage values are with respect to ground terminal.
5. F maximum: $t_r = t_f \leq 6$ ns.



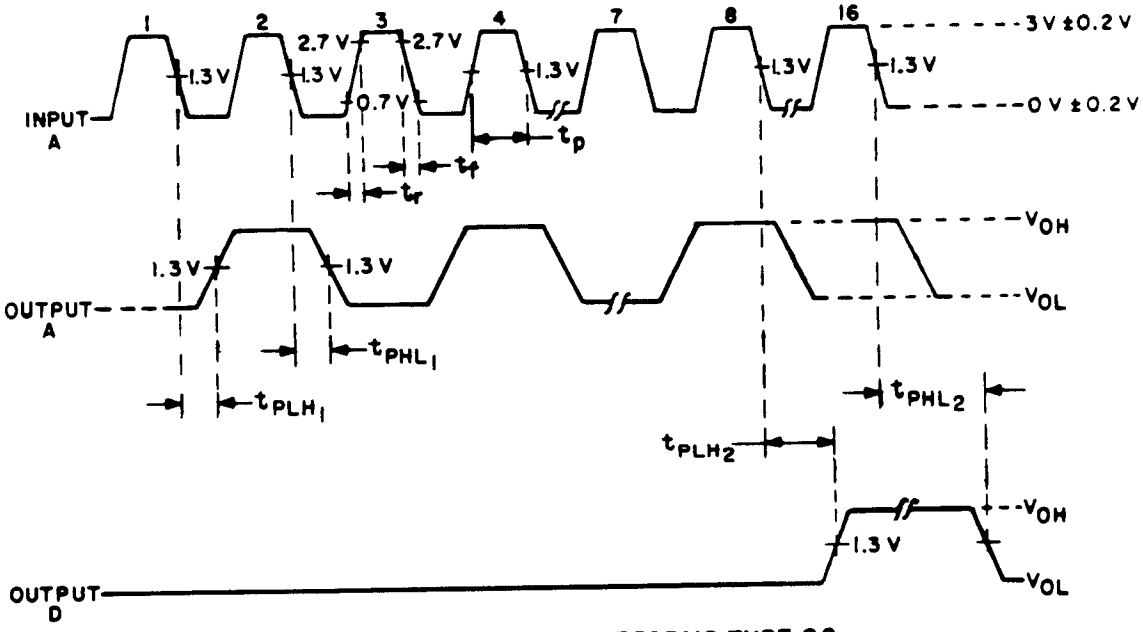
CLEAR VOLTAGE WAVEFORMS TYPES 01,02

FIGURE 5. Switching time test circuit and waveforms for device types 01 and 02.

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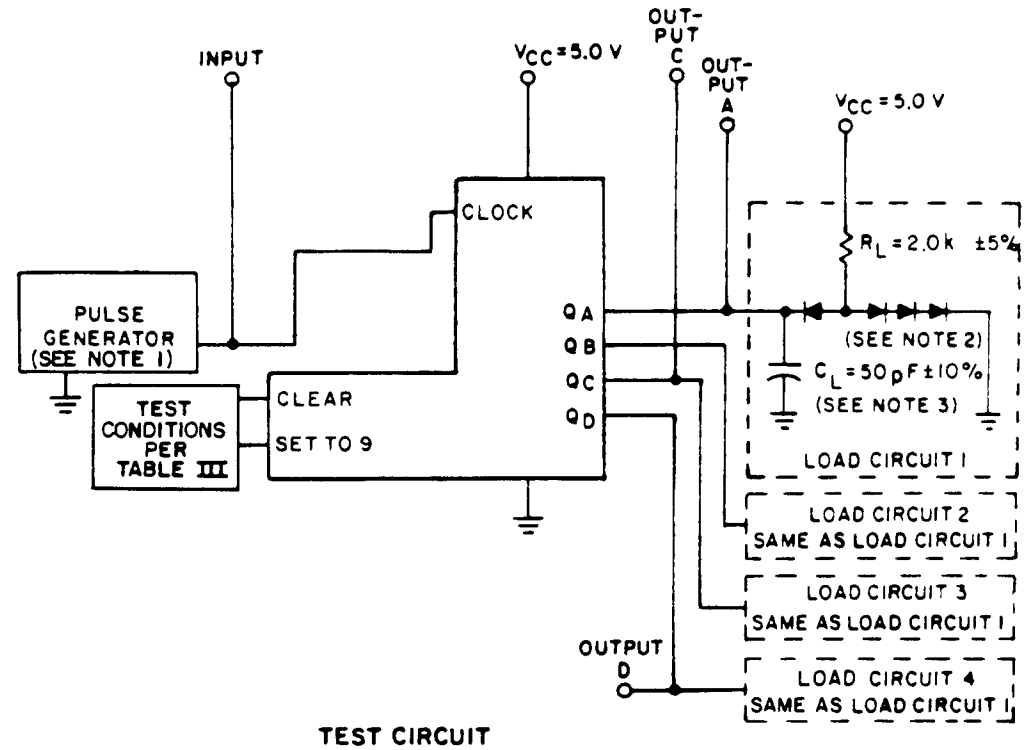
COUNT VOLTAGE WAVEFORMS, TYPE 01



COUNT VOLTAGE WAVEFORMS, TYPE 02

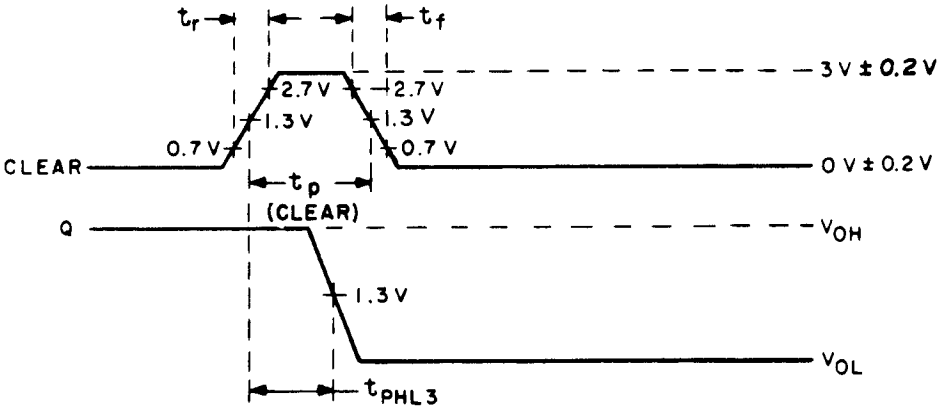
FIGURE 5. Switching time test circuit and waveforms for device types 01 and 02 - Continued.

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TEST CIRCUIT

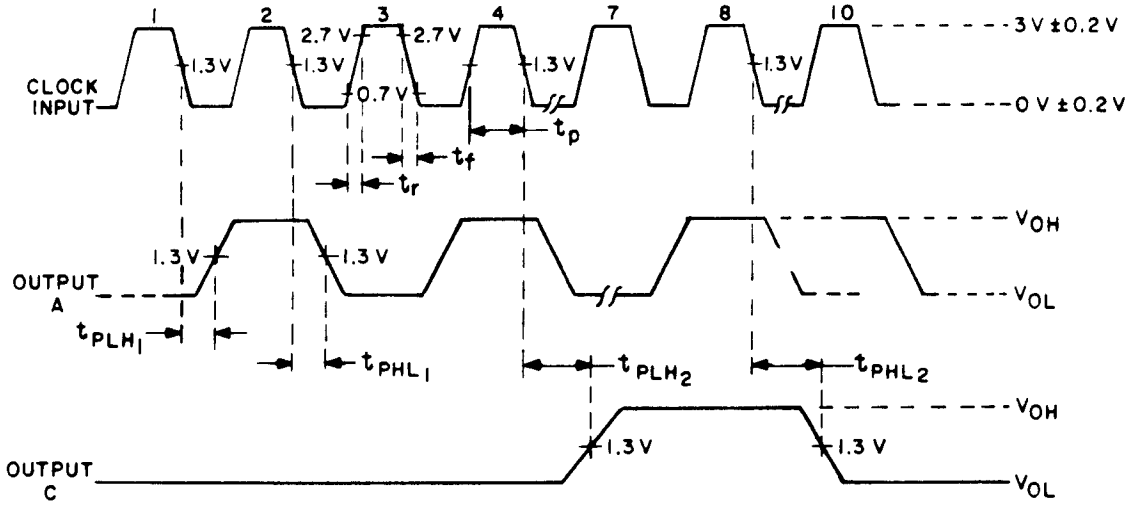
NOTES:
1. The pulse generator has the following characteristics: V_{gen} = 3 V, $t_r \leq 15$ ns, $t_f \leq 6$ ns, t_p = .5 µs, PRR ≤ 1 MHz, $Z_{out} \approx 50$, $t_{p(clear)} \geq 20$ ns.
2. All diodes are 1N3064 or equivalent.
3. C_L includes probe and jig capacitance.
4. Voltage values are with respect to ground terminal.
5. F maximum: $t_r = t_f \leq 6$ ns.



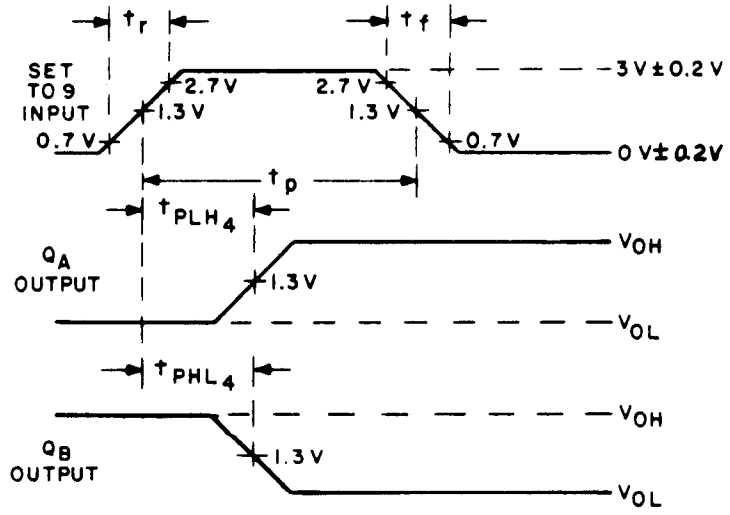
CLEAR VOLTAGE WAVEFORMS

FIGURE 6. Switching time test circuit and waveforms for device type 03.

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COUNT VOLTAGE WAVEFORMS



SET-TO-NINE VOLTAGE WAVEFORMS

FIGURE 6. Switching time test circuit and waveforms for device type 03 - Continued.

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TABLE III. Group A inspection for device type 01.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.7 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E,F Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Test limits Min	Test limits Max	Unit
				1A	1CLR	1QA	1B	1QB	1QC	1QD	GND	2QD	2QC	2QB	2B	2QA	2CLR	2A	V_{CC}				
1 T_C = 25°C	V_{OH}	3006	1	1/	2/	-0.4 mA					GND								4.5 V	1QA	2.5		V
	"	"	2		"		1/	-0.4 mA			"								"	1QB	"		"
	"	"	3		"		3/		-0.4 mA		"								"	1QC	"		"
	"	"	4		"		4/			-0.4 mA	"								"	1QD	"		"
	"	"	5								"			-0.4 mA	1/	-0.4 mA	2/	1/	"	2QA	"		"
	"	"	6								"		-0.4 mA		3/		"		"	2QB	"		"
	"	"	7								"				4/		"		"	2QC	"		"
	"	"	8								"	-0.4 mA							"	2QD	"		"
	V_{OL}	3007	9		2.0 V	5/					"								"	1QA		0.4	"
	"	"	10		"			4 mA			"								"	1QB		"	"
	"	"	11		"				4 mA		"								"	1QC		"	"
	"	"	12							4 mA	"								"	1QD		"	"
	"	"	13								"					5/			"	2QA		"	"
	"	"	14								"			4 mA			2.0 V		"	2QB		"	"
	"	"	15								"		4 mA				"		"	2QC		"	"
	"	"	16								"	4 mA					"		"	2QD		"	"
	V_{IC}		17	-18 mA							"								"	1A		-1.5	"
	"		18		-18 mA						"								"	1CLR		"	"
	"		19				-18 mA				"								"	1B		"	"
	"		20								"				-18 mA				"	2B		"	"
	"		21								"						-18 mA		"	2CLR		"	"
	"		22								"							-18 mA	"	2A		"	"
	I_{IL1}	3009	23		0.4 V						"								5.5 V	1CLR	6/	6/	6/
	I_{IL2}	"	24	0.4 V	GND						"								"	1A	"	"	"
	I_{IL3}	"	25		GND		0.4 V				"								"	1B	"	"	"
	I_{IL1}	"	26								"								"	2CLR	"	"	"
	I_{IL2}	"	27								"						0.4 V		"	2A	"	"	"
	I_{IL3}	"	28								"					0.4 V	GND	0.4 V	"	2B	"	"	"
	I_{IH1}	3010	29		2.7 V						"								"	1CLR		20	µA
	I_{IH2}	"	30	2.7 V							"								"	1A		100	"
	I_{IH3}	"	31				2.7 V				"								"	1B		200	"
	I_{IH1}	"	32								"						2.7 V		"	2CLR		20	"
	I_{IH2}	"	33								"							2.7 V	"	2A		100	"
	I_{IH3}	"	34								"				2.7 V				"	2B		200	"
	I_{IH4}	"	35		5.5 V						"								"	1CLR		100	"
	I_{IH5}	"	36	5.5 V							"								"	1A		200	"
	I_{IH6}	"	37				5.5 V				"								"	1B		400	"
	I_{IH4}	"	38								"						5.5 V		"	2CLR		100	"
	I_{IH5}	"	39								"							5.5 V	"	2A		200	"
	I_{IH6}	"	40								"				5.5 V				"	2B		400	"
	I_{OS}	3011	41		1/	2/	GND													1QA	-15	-100	mA
	"	"	42		"	"		1/	GND										"	1QB	"	"	"
	"	"	43		"	"		3/		GND									"	1QC	"	"	"
	"	"	44		"	"		4/			GND								"	1QD	"	"	"
	"	"	45								"					GND			"	2QA	"	"	"
	"	"	46								"				1/		2/		"	2QB	"	"	"
	"	"	47								"				3/		"		"	2QC	"	"	"
	"	"	48								"	GND	GND	GND	4/		"	1/	"	2QD	"	"	"
	I_{CC}	3005	49	GND	2/	GND					"				GND		"	GND	"	VCC		26	"

S notes at end of device type 01.

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may may be high ≥2.0 V, low ≤0.7 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E,F Test no.	1 1A	2 1CLR	3 1QA	4 1B	5 1QB	6 1QC	7 1QD	8 GND	9 2QD	10 2QC	11 2QB	12 2B	13 2QA	14 2CLR	15 2A	16 V_{CC}	Measured terminal	Test limits Min	Test limits Max	Unit
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = 125°C and V_{IC} tests are omitted.																						
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																						
7 7/ T_C = 25°C	Truth table tests	3014	50 8/	A	A	L	A	L	L	L	GND	L	L	L	A	L	A	A	5.0 V	9/			
	"	"	51	A	"	"	b	"	"	"	"	"	"	"	B	"	"	A	"				
	"	"	52	B	"	"	b	"	"	"	"	"	"	"	A	"	"	B	"				
	"	"	53	"	"	"	A	"	"	"	"	"	"	"	A	"	B	"	"				
	"	"	54	"	B	"	A	"	"	"	"	"	"	"	B	"	"	"	"				
	"	"	55	"	"	"	B	H	"	"	"	"	"	H	A	"	"	"	"				
	"	"	56	"	"	"	A	H	"	"	"	"	"	H	B	"	"	"	"				
	"	"	57	"	"	"	B	L	"	"	"	"	"	L	A	"	"	"	"				
	"	"	58	"	"	"	A	L	"	"	"	"	"	"	B	"	"	"	"				
	"	"	59	"	"	"	B	H	"	"	"	"	"	"	A	"	"	"	"				
	"	"	60	"	"	"	A	H	"	"	"	"	"	"	B	"	"	"	"				
	"	"	61	"	"	"	B	L	"	H	"	"	H	"	A	"	"	"	"				
	"	"	62	"	"	"	A	L	"	"	"	"	"	"	B	"	"	"	"				
	"	"	63	"	"	"	B	H	"	"	"	"	"	"	A	"	"	"	"				
	"	"	64	A	"	"	A	H	"	"	"	"	"	"	B	"	"	"	"				
	"	"	65	B	"	H	B	L	"	"	"	"	"	"	A	"	H	"	"				
8 7/	Same tests, terminal conditions, and limits as for subgroup 7, except T_C = 125°C and -55°C.																						
9 T_C = 25°C	F_{MAX1} 10/	3003	66	IN	GND	OUT					GND						OUT	GND	IN	5.0 V	1A > 1QA	25		MHz
	F_{MAX1} 10/	Fig.	67								"									"	2A > 2QA	25		"
	F_{MAX2} 10/	5	68		GND		IN			OUT	"									"	1B > 1QD	12.5		"
	F_{MAX2} 10/	"	69								"	OUT			IN			GND		"	2B > 2QD	12.5		"
	t_{PLH1}	"	70	IN	GND				OUT		"									"	1A > 1QC	2	65	ns
	t_{PLH1}	"	71								"		OUT					GND	IN	"	2A > 2QC	"		"
	t_{PHL1}	"	72	IN	GND				OUT		"									"	1A > 1QC	"		"
	t_{PHL1}	"	73								"		OUT					GND	IN	"	2A > 2QC	"		"
	t_{PLH2}	"	74		GND		IN			OUT	"									"	1B > 1QD	"	26	"
	t_{PLH2}	"	75								"	OUT			IN			GND		"	2B > 2QD	"		"
	t_{PLH2}	"	76		GND		IN			OUT	"									"	1B > 1QD	"		"
	t_{PHL2}	"	77								"	OUT			IN			GND		"	2B > 2QD	"		"
	t_{PHL2}	"	78	11/	IN	OUT					"									"	1CLR > 1QA	"	44	"
	t_{PHL3}	"	79		"		11/	OUT			"									"	1CLR > 1QB	"		"
	"	"	80				"		OUT		"									"	1CLR > 1QC	"		"
	"	"	81							OUT	"									"	1CLR > 1QD	"		"
	"	"	82								"						OUT	IN	11/	"	2CLR > 2QA	"		"
	"	"	83								"			OUT	11/				"	"	2CLR > 2QB	"		"
	"	"	84								"		OUT		"				"	"	2CLR > 2QC	"		"
	"	"	85								"	OUT			"				"	"	2CLR > 2QD	"		"
10 T_C = 125°C	F_{MAX1} F_{MAX2} t_{PLH1} t_{PHL1} t_{PLH2} t_{PHL2} t_{PHL3}	Same tests and terminal conditions as for subgroup 9, except T_C = 125°C.																			25	84	MHz
																					12.5	84	MHz
																					2	34	ns
																					"	34	"
																					"	56	"

See footnotes at end of device type 01.

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may may be high ≥2.0 V, low ≤0.7 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E,F Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Test limits Min	Test limits Max	Unit
				1A	1CLR	1QA	1B	1QB	1QC	1QD	GND	2QD	2QC	2QB	2B	2QA	2CLR	2A	V_{CC}				
11 T_C = -55°C		Same tests, terminal conditions, and limits as for subgroup 10, except T_C = -55°C.																					

1/ Apply ⎍ 2.0 V min./5.5 V max. 0.0 ±0.2 V prior to test after clear pulse.

2/ Apply ⎍ 2.0 V min./5.5 V max. 0.0 ±0.2 V pulse prior to test.

3/ Apply 2 pulses prior to test after clear pulse (see 1/).

4/ Apply 4 pulses prior to test after clear pulse (see 1/).

5/ For tests 9 and 13, I_{IL3} maximum value plus 4 mA shall be applied to output Q_A.

6/ I_{IL} limits shall be as follow:

Test	Min./Max. limits (mA) Circuit A	Circuit E	Circuit B
I_{IL1}	-.15/-.38	-.12/-.36	-.16/-.40
I_{IL2}	-.35/-1.6	-1.0/-2.4	-.35/-1.6
I_{IL3}	-.60/-2.4	-1.3/-3.2	-.60/-2.4

7/ Only a summary of attributes data is required.

8/ A = 2.0 V min. and B = 0.7 V max.

9/ Output voltages shall be either:
a. H ≥2.5 V and L ≤0.4 V when using a high speed double comparator, or
b. H ≥1.5 V and L ≤1.5 V when using a high speed single comparator.

10/ F_{MAX1} and F_{MAX2} minimum limits specified are the frequency of the input pulse.
The output pulse shall be one-half the input frequency when measuring Q_A.
The output shall be one-fifth of the input frequency when measuring Q_D.

11/ Apply sufficient ⎍ 2.0 V min./5.5 V max. 0.0 ±0.2 V pulses to set output high prior to test.

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TABLE III. Group A inspection for device type 02.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.7 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases A,B,C,D Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Test limits Min	Test limits Max	Unit
				1A	1CLR	1QA	1QB	1QC	1QD	GND	2QD	2QC	2QB	2QA	2CLR	2A	V_{CC}				
1 T_C = 25°C	V_{OL}	3007	1		2.0 V	4 mA				GND							4.5 V	1QA		0.4	V
	"	"	2		"		4 mA			"							"	1QB		"	"
	"	"	3		"			4 mA		"							"	1QC		"	"
	"	"	4		"				4 mA	"							"	1QD		"	"
	"	"	5		"					"				4 mA			"	2QA		"	"
	"	"	6		"					"			4 mA		2.0 V		"	2QB		"	"
	"	"	7		"					"		4 mA			"		"	2QC		"	"
	"	"	8		"					"	4 mA				"		"	2QD		"	"
	V_{OH}	3006	9	1/	2/	-400 µA	-400 µA	-400 µA	-400 µA	"				-400 µA	2/	1/	"	1QA	2.5		"
	"	"	10	"	"					"					"	"	"	1QB	"		"
	"	"	11	"	"					"					"	"	"	1QC	"		"
	"	"	12		"					"					"	"	"	1QD	"		"
	"	"	13							"					"	"	"	2QA	"		"
	"	"	14							"			-400 µA		"	"	"	2QB	"		"
	"	"	15							"		-400 µA			"	"	"	2QC	"		"
	"	"	16							"	-400 µA				"	"	"	2QD	"		"
	V_{IC}		17	-18 mA						"							"	1A		-1.5	"
	"		18		-18 mA					"							"	1CLR		"	"
	"		19							"						-18 mA	"	2A		"	"
	"		20							"					-18 mA		"	2CLR		"	"
	I_{IL1}	3009	21		0.4 V					"							5.5 V	1CLR	3/	3/	3/
	I_{IL1}	"	22							"					0.4 V		"	2CLR	"	"	"
	I_{IL2}	"	23	0.4 V	GND					"							"	1A	"	"	"
	I_{IL2}	"	24							"					GND	0.4 V	"	2A	"	"	"
	I_{IH1}	3010	25		2.7 V					"							"	1CLR		20	µA
	I_{IH1}	"	26							"					2.7 V		"	2CLR		20	"
	I_{IH2}	"	27	2.7 V						"							"	1A		100	"
	I_{IH2}	"	28							"						2.7 V	"	2A		100	"
	I_{IH4}	"	29		5.5 V					"							"	1CLR		100	"
	I_{IH4}	"	30							"					5.5 V		"	2CLR		100	"
	I_{IH5}	"	31	5.5 V						"							"	1A		400	"
	I_{IH5}	"	32							"						5.5 V	"	2A		400	"
	I_{OS}	3011	33	1/	2/	GND	GND	GND	GND	"				GND	2/	1/	"	1QA	-15	-100	mA
	"	"	34	"	"					"					"	"	"	1QB	"	"	"
	"	"	35	"	"					"					"	"	"	1QC	"	"	"
	"	"	36	"	"					"					"	"	"	1QD	"	"	"
	"	"	37							"					"	"	"	2QA	"	"	"
	"	"	38							"			GND		"	"	"	2QB	"	"	"
	"	"	39							"		GND			"	"	"	2QC	"	"	"
	"	"	40							"	GND				"	"	"	2QD	"	"	"
	I_{CC}	3005	41	GND	2/					"					"	GND	"	VCC		26	"
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = 125°C and V_{IC} tests are omitted.																				
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																				

See footnotes at end of device type 02.

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TABLE III. Group A inspection for device type 02 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.7 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases A,B,C,D Test no.	1 1A	2 1CLR	3 1QA	4 1QB	5 1QC	6 1QD	7 GND	8 2QD	9 2QC	10 2QB	11 2QA	12 2CLR	13 2A	14 V_{CC}	Measured terminal	Test limits Min	Test limits Max	Unit
7 4/ T_C = 25°C	Truth table tests	3014	42	A	A	L	L	L	L	GND	L	L	L	L	A	A	5.0 V	5/			
	"	"	43	B	A	"	"	"	"	"	"	"	"	"	A	B	"				
	"	"	44	B	B	"	"	"	"	"	"	"	"	"	B	B	"				
	"	"	45	A	"	"	"	"	"	"	"	"	"	"	"	A	"				
	"	"	46	B	"	H	"	"	"	"	"	"	"	H	"	B	"				
	"	"	47	A	"	H	"	"	"	"	"	"	"	H	"	A	"				
	"	"	48	B	"	L	H	"	"	"	"	"	H	L	"	B	"				
	"	"	49	A	"	L	"	"	"	"	"	"	"	L	"	A	"				
	"	"	50	B	"	H	"	"	"	"	"	"	"	H	"	B	"				
	"	"	51	A	"	H	"	"	"	"	"	"	"	H	"	A	"				
	"	"	52	B	"	L	L	H	"	"	"	H	L	L	"	B	"				
	"	"	53	A	"	L	"	"	"	"	"	"	"	L	"	A	"				
	"	"	54	B	"	H	"	"	"	"	"	"	"	H	"	B	"				
	"	"	55	A	"	H	"	"	"	"	"	"	"	H	"	A	"				
	"	"	56	B	"	L	H	"	"	"	"	"	H	L	"	B	"				
	"	"	57	A	"	L	"	"	"	"	"	"	"	L	"	A	"				
	"	"	58	B	"	H	"	"	"	"	"	"	"	H	"	B	"				
	"	"	59	A	"	H	"	"	"	"	"	"	"	H	"	A	"				
	"	"	60	B	"	L	L	L	H	"	H	L	L	L	"	B	"				
	"	"	61	A	"	L	"	"	"	"	"	"	"	L	"	A	"				
	"	"	62	B	"	H	"	"	"	"	"	"	"	H	"	B	"				
	"	"	63	A	"	H	"	"	"	"	"	"	"	H	"	A	"				
	"	"	64	B	"	L	H	"	"	"	"	"	H	L	"	B	"				
	"	"	65	A	"	L	"	"	"	"	"	"	"	L	"	A	"				
	"	"	66	B	"	H	"	"	"	"	"	"	"	H	"	B	"				
	"	"	67	A	"	H	"	"	"	"	"	"	"	H	"	A	"				
	"	"	68	B	"	L	L	H	"	"	"	H	L	L	"	B	"				
	"	"	69	A	"	L	"	"	"	"	"	"	"	L	"	A	"				
	"	"	70	B	"	H	"	"	"	"	"	"	"	H	"	B	"				
	"	"	71	A	"	H	"	"	"	"	"	"	"	H	"	A	"				
	"	"	72	B	"	L	H	"	"	"	"	"	H	L	"	B	"				
	"	"	73	A	"	L	"	"	"	"	"	"	"	L	"	A	"				
	"	"	74	B	"	H	"	"	"	"	"	"	"	H	"	B	"				
	"	"	75	B	A	L	L	L	L	"	L	L	L	L	A	B	"				
8	Same tests, terminal conditions, and limits as for subgroup 7, except T_C = 125°C and T_C = -55°C.																				
9 T_C = 25°C	F_{MAX1} 6/	3003	76	IN	GND	OUT				GND							5.0 V	1A ▸ 1QA	25		MHz
	F_{MAX1} 6/	Fig.	77							"				OUT	GND	IN	"	2A ▸ 2QA	25		MHz
	t_{PLH1}	5	78	IN	GND	OUT				"							"	1A ▸ 1QA	2	25	ns
	t_{PLH1}	"	79							"				OUT	GND	IN	"	2A ▸ 2QA	"	"	"
	t_{PHL1}	"	80	IN	GND	OUT				"							"	1A ▸ 1QA	"	"	"
	t_{PHL1}	"	81							"				OUT	GND	IN	"	2A ▸ 2QA	"	"	"
	t_{PLH2}	"	82	IN	GND				OUT	"							"	1A ▸ 1QD	"	65	"
	t_{PLH2}	"	83							"	OUT				GND	IN	"	2A ▸ 2QD	"	"	"
	t_{PHL2}	"	84	IN	GND				OUT	"							"	1A ▸ 1QD	"	"	"
	t_{PHL2}	"	85							"	OUT				GND	IN	"	2A ▸ 2QD	"	"	"
	t_{PHL3}	"	86	7/	IN	OUT				"							"	1CLR ▸ 1QA	"	44	"
	"	"	87	"	"		OUT			"							"	1CLR ▸ 1QB	"	"	"
	"	"	88	"	"			OUT		"							"	1CLR ▸ 1QC	"	"	"
	"	"	89	"	"				OUT	"							"	1CLR ▸ 1QD	"	"	"
	"	"	90							"				OUT	IN	7/	"	2CLR ▸ 2QA	"	"	"
	"	"	91							"			OUT		"	"	"	2CLR ▸ 2QB	"	"	"
	"	"	92							"		OUT			"	"	"	2CLR ▸ 2QC	"	"	"
	"	"	93							"	OUT				"	"	"	2CLR ▸ 2QD	"	"	"
10 T_C = 125°C	F_{MAX1}	Same tests and terminal conditions as for subgroup 9, except T_C = 125°C.																	25		MHz
	t_{PLH1}																		2	33	ns
	t_{PHL1}																		"	33	"
	t_{PLH2}																		"	84	"
	t_{PHL2}																		"	84	"
	t_{PHL3}																		"	56	"

See footnotes at end of device type 02.

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TABLE III. Group A inspection for device type 02 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.7 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases A,B,C,D Test no.	1 1A	2 1CLR	3 1QA	4 1QB	5 1QC	6 1QD	7 GND	8 2QD	9 2QC	10 2QB	11 2QA	12 2CLR	13 2A	14 V_{CC}	Measured terminal	Test limits Min	Test limits Max	Unit
11 T_C = -55°C		Same tests, terminal conditions, and limits as for subgroup 10 except T_C = 55°C.																			

1/ Apply ⎍⎍ 2.0 V min./5.5 V max. / 0.0 ±0.2 V to clear input and then to the A input sufficient times prior to test to set the output high.

2/ Apply ⎍⎍ 2.0 V min./5.5 V max. / 0.0 ±0.2 V prior to test.

3/ I_{IL} limits shall be as follows:

Test	Min./Max. limits (mA) Circuits A	E	B	D
I_{IL1}	-.15/-.38	-.12/-.36	-.16/-.40	-.12/-.36
I_{IL2}	-.35/-1.6	-1.0/-2.4	-.35/-1.60	-.35/-1.60

4/ A ≥2.0 V and B ≤0.7 V.

5/ Output voltages shall be either:
a. H ≥2.5 V and L ≤0.4 V max when using a high speed checker double comparator, or
b. H ≥1.5 V and L ≤1.5 V when using a high speed checker single comparator.

6/ F_{MAX} minimum limit specified is the frequency of the input pulse. The output pulse shall be one-half the input frequency.

7/ Apply sufficient ⎍⎍ 2.0 V min./5.5 V max. / 0.0 V ±0.2 V pulses to set outputs high prior to test.

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TABLE III. Group A inspection for device type 03.
Terminal conditions (pins not designated may may be high ≥2.0 V, low ≤0.7 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E,F Test no.	1 1CLK	2 1CLR	3 1QA	4 1ST9	5 1QB	6 1QC	7 1QD	8 GND	9 2QD	10 2QC	11 2QB	12 2ST9	13 2QA	14 2CLK	15 2CLK	16 V_{CC}	Measured terminal	Test Limits Min	Test Limits Max	Unit
1 T_C = 25°C	V_{OH}	3006	1	1/ 3/ 4/	2/	-0.4 mA	0.7 V	-0.4 mA			GND				0.7 V	-0.4 mA	2/	1/ 3/ 4/	4.5 V	1QA	2.5		V
	"	"	2		"		"		-0.4 mA		"				"		"		"	1QB	"		"
	"	"	3		"		"			-0.4 mA	"				"		"		"	1QC	"		"
	"	"	4		0.7 V		2.0 V				"				"		"		"	1QD	"		"
	"	"	5								"				"		"		"	2QA	"		"
	"	"	6								"			-0.4 mA	"		"		"	2QB	"		"
	"	"	7								"		-0.4 mA		"		"		"	2QC	"		"
	"	"	8								"	-0.4 mA			2.0 V		0.7 V		"	2QD	"		"
	V_{OL}	3007	9		2.0 V	4 mA	0.7 V	4 mA			"								"	1QA		0.4	"
	"	"	10		"		"		4 mA		"								"	1QB		"	"
	"	"	11		"		"			4 mA	"								"	1QC		"	"
	"	"	12		"		"				"								"	1QD		"	"
	"	"	13								"			4 mA	0.7 V	4 mA	2.0 V		"	2QA		"	"
	"	"	14								"		4 mA		"		"		"	2QB		"	"
	"	"	15								"	4 mA			"		"		"	2QC		"	"
	"	"	16								"				"		"		"	2QD		"	"
	V_{IC}		17	-18 mA							"								"	1CLK		1.5	"
	"		18		-18 mA						"								"	1CLR		"	"
	"		19				-18 mA				"								"	1ST9		"	"
	"		20								"				-18 mA				"	2ST9		"	"
	"		21								"						-18 mA		"	2CLR		"	"
	"		22								"							-18 mA	"	2CLK		"	"
	I_{IL1}	3009	23		0.4 V						"								5.5 V	1CLR	5/	5/	5/
	I_{IL2}	"	24	0.4 V	GND		GND				"								"	1CLK	"	"	"
	I_{IL1}	"	25				0.4 V				"								"	1ST9	"	"	"
	I_{IL1}	"	26								"						0.4 V		"	2CLR	"	"	"
	I_{IL2}	"	27								"				GND		GND	0.4 V	"	2CLK	"	"	"
	I_{IL1}	"	28								"				0.4 V				"	2ST9	"	"	"
	I_{IH1}	3010	29		2.7 V						"								"	1CLR		20	µA
	I_{IH2}	"	30	2.7 V							"								"	1CLK		100	"
	I_{IH1}	"	31				2.7 V				"								"	1ST9		20	"
	I_{IH1}	"	32								"						2.7 V		"	2CLR		20	"
	I_{IH2}	"	33								"							2.7 V	"	2CLK		100	"
	I_{IH1}	"	34								"				2.7 V				"	2ST9		20	"
	I_{IH4}	"	35		5.5 V						"								"	1CLR		100	"
	I_{IH5}	"	36	5.5 V							"								"	1CLK		200	"
	I_{IH4}	"	37				5.5 V				"								"	1ST9		100	"
	I_{IH4}	"	38								"						5.5 V		"	2CLR		100	"
	I_{IH5}	"	39								"							5.5 V	"	2CLK		200	"
	I_{IH4}	"	40								"				5.5 V				"	2ST9		100	"
	I_{OS}	3011	41	1/ 3/ 4/	2/	GND	GND	GND			"								"	1QA	-15	-100	mA
	"	"	42		"		"				"								"	1QB	"	"	"
	"	"	43		"		"		GND		"								"	1QC	"	"	"
	"	"	44		GND		5.5 V			GND	"								"	1QD	"	"	"
	"	"	45								"				GND	GND			"	2QA	"	"	"
	"	"	46								"			GND	"				"	2QB	"	"	"
	"	"	47								"		GND		"				"	2QC	"	"	"
	"	"	48								"	GND			5.5 V				"	2QD	"	"	"
	I_{CC}	3005	49	GND	2/		GND				"				GND		2/	GND	"	VCC		26	"

See fr ›s at end of device type 03.

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TABLE III. Group A inspection for device type 03 - Continued.
Terminal conditions (pins not designated may may be high ≥2.0 V, low ≤0.7 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E,F Test no.	1 1CLK	2 1CLR	3 1QA	4 1ST9	5 1QB	6 1QC	7 1QD	8 GND	9 2QD	10 2QC	11 2QB	12 2ST9	13 2QA	14 2CLK	15 2CLK	16 V_{CC}	Measured terminal	Test limits Min	Test limits Max	Unit
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = 125°C and V_{IC} tests are omitted.																						
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																						
7 6/ T_C = 25°C	Truth table tests	3014	50 7/	A	A	L	B	L	L	L	GND	L	L	L	B	L	A	A	5.0 V			8/	
	"	"	51	B	A	"	"	"	"	"	"	"	"	"	"	"	A	B	"				
	"	"	52	A	B	"	"	"	"	"	"	"	"	"	"	"	B	A	"				
	"	"	53	B	"	H	"	"	"	"	"	"	"	"	"	H	"	B	"				
	"	"	54	A	"	H	"	"	"	"	"	"	"	"	"	H	"	A	"				
	"	"	55	B	"	L	H	"	"	"	"	"	"	H	"	L	"	B	"				
	"	"	56	A	"	L	"	"	"	"	"	"	"	"	"	L	"	A	"				
	"	"	57	B	"	H	"	"	"	"	"	"	"	"	"	H	"	B	"				
	"	"	58	A	"	H	"	"	"	"	"	"	"	"	"	H	"	A	"				
	"	"	59	B	"	L	"	L	H	"	"	"	H	L	"	L	"	B	"				
	"	"	60	A	"	L	"	"	"	"	"	"	"	"	"	L	"	A	"				
	"	"	61	B	"	H	"	"	"	"	"	"	"	"	"	H	"	B	"				
	"	"	62	A	"	H	"	"	"	"	"	"	"	"	"	H	"	A	"				
	"	"	63	B	"	L	H	"	"	"	"	"	"	H	"	L	"	B	"				
	"	"	64	A	"	L	"	"	"	"	"	"	"	"	"	L	"	A	"				
	"	"	65	B	"	H	"	"	"	"	"	"	"	"	"	H	"	B	"				
	"	"	66	A	"	H	"	"	"	"	"	"	"	"	"	H	"	A	"				
	"	"	67	B	"	L	"	L	L	H	"	H	L	L	"	L	"	B	"				
	"	"	68	A	"	L	"	"	"	"	"	"	"	"	"	L	"	A	"				
	"	"	69	B	"	H	"	"	"	"	"	"	"	"	"	H	"	B	"				
	"	"	70	A	A	L	"	"	"	L	"	L	"	"	"	L	A	A	"				
	"	"	71	"	B	L	"	"	"	L	"	L	"	"	"	L	B	"	"				
	"	"	72	"	B	H	A	"	"	H	"	H	"	"	A	H	B	"	"				
8 6/	Same tests, terminal conditions, and limits as for subgroup 7, except T_C = 125°C and -55°C.																						
9 T_C = 25°C	F_{MAX1} 9/	3003	73	IN	GND	OUT	GND				GND				GND	OUT	GND	IN	5.0 V	CLK » 1QA	25		MHz
	F_{MAX1} 9/	Fig.	74								"								"	CLK » 2QA	25		"
	t_{PLH4}	6	75		2/	OUT	IN				"								"	ST9 » 1QA	2	44	ns
	t_{PLH4}	"	76								"				IN	OUT	2/		"	ST9 » 2QA	2	44	"
	t_{PLH1}	"	77	IN	GND	OUT	GND				"								"	CLK » 1QA	25		"
	t_{PLH1}	"	78								"				GND	OUT	GND	IN	"	CLK » 2QA	"	"	"
	t_{PHL1}	"	79	IN	GND	OUT	GND				"								"	CLK » 1QA	"	"	"
	t_{PHL1}	"	80								"				GND	OUT	GND	IN	"	CLK » 2QA	"	"	"
	t_{PLH2}	"	81	IN	GND		GND		OUT		"								"	CLK » 1QC	"	59	"
	t_{PLH2}	"	82								"		OUT		GND		GND	IN	"	CLK » 2QC	"	"	"
	t_{PLH2}	"	83	IN	GND		GND		OUT		"								"	CLK » 1QC	"	"	"
	t_{PHL2}	"	84								"		OUT		GND		GND	IN	"	CLK » 2QC	"	"	"
	t_{PHL3}	"	85	10/	IN	OUT	GND				"								"	CLK » 1QA	"	44	"
	"	"	86	"	"		"	OUT			"								"	CLR » 1QB	"	"	"
	"	"	87	"	"		"		OUT		"								"	CLR » 1QC	"	"	"
	"	"	88	"	"		"			OUT	"								"	CLR » 1QD	"	"	"
	"	"	89								"				GND	OUT	IN	10/	"	CLR » 2QA	"	"	"
	"	"	90								"			OUT	"		"	"	"	CLR » 2QB	"	"	"
	"	"	91								"		OUT		"		"	"	"	CLR » 2QC	"	"	"
	"	"	92								"	OUT			"		"	"	"	CLR » 2QD	"	"	"
	t_{PHL4}	"	93	10/	GND		IN	OUT			"								"	ST9 » 1QB	"	41	"
	t_{PHL4}	"	94								"			OUT	IN		GND	10/	"	ST9 » 2QB	"	41	"
10 T_C = 125°C	F_{MAX1}																				25		MHz
	t_{PLH4}																				2	57	ns
	t_{PLH1}																				"	33	"
	t_{PHL1}																				"	33	"
	t_{PLH2}																				"	77	"
	t_{PHL2}																				"	77	"
	t_{PHL3}																				"	56	"
	t_{PHL4}																				"	53	"
	Same tests and terminal conditions as for subgroup 9, except T_C = 125°C.																						

See footnotes at end of device type 03.

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TABLE III. Group A inspection for device type 03 - Continued.
Terminal conditions (pins not designated may may be high ≥2.0 V, low ≤0.7 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E,F Test no.	1 1CLK	2 1CLR	3 1QA	4 1ST9	5 1QB	6 1QC	7 1QD	8 GND	9 2QD	10 2QC	11 2QB	12 2ST9	13 2QA	14 2CLR	15 2CLK	16 V_{CC}	Measured terminal	Test limits Min	Test limits Max	Unit
11 T_C = -55°C			Same tests, terminal conditions, and limits as for subgroup 10, except T_C = -55°C.																				

1/ Apply ⎍ 2.0 V min./5.5 V max. 0.0 ±0.2 V prior to test after clear pulse.

2/ Apply ⎍ 2.0 V min./5.5 V max. 0.0 ±0.2 V pulse prior to test.

3/ Apply 2 pulses after clear pulse (see 1/).

4/ Apply 4 pulses after clear pulse (see 1/).

5/ I_{IL} limits shall be as follows:

Test	Min./Max. limits (mA) Circuit A	Circuit E	Circuit B
I_{IL1} (CLR)	-.15/-.38	-.135/-.37	-.16/-.40
I_{IL1} (ST9)	-.16/-.4	-.135/-.37	-.16/-.40
I_{IL2}	-.35/-1.6	-1.0/-2.4	-.35/-1.6

6/ Only a summary of attributes data is required.

7/ A = 2.0 V min. and B = 0.7 V max.

8/ Output voltages shall be either:
a. H ≥2.5 V and L ≤0.4 V when using a high speed double comparator, or
b. H ≥1.5 V and L ≤1.5 V when using a high speed single comparator.

9/ F_{MAX1} minimum limit specified is the frequency of the input pulse. The output pulse shall be one-half the input frequency.

10/ Apply sufficient | | 2.0 V min./5.5 V max. 0.0 ±0.2 V pulses to set output high prior to test.

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4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspections. The following additional criteria shall apply:

a. Burn-in test (method 1015 of MIL-STD-883).

(1) Test condition D or E, using the circuit shown on figure 4, or equivalent.

(2) T_A = 125°C minimum.

b. Interim and final electrical test parameters shall be as specified in table II, except interim electrical parameters test prior to burn-in is optional at the discretion of the manufacturer.

c. The percent defective allowable (PDA) for class S devices shall be as specified in MIL-M-38510. The PDA for class B devices shall be 10 percent based on failures from group A, subgroup 1 tests after cooldown as the final electrical test in accordance with method 5004 of MIL-STD-883, and with no intervening electrical measurements. If interim electrical parameter tests are performed prior to burn-in, failures resulting from pre burn-in screening may be excluded from the PDA. If interim electrical parameter tests prior to burn-in are omitted, then all screening failures shall be included in the PDA. The verified failures of group A, subgroup 1 after burn-in divided by the total number of devices submitted for burn-in in that lot shall be used to determine the percent defective for that lot, and the lot shall be accepted or rejected based on the PDA for the applicable device class.

4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Tests shall be as specified in table II herein.

b. Subgroups 4, 5, and 6 of table I of method 5005 of MIL-STD-883 shall be omitted.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883. Electrical tests shall be as specified in table II herein.

4.4.3 Group C inspection. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II herein.

b. Subgroups 3 and 4 shall be added to group C inspection for class C devices and shall consist of the tests, conditions, and limits specified for subgroups 10 and 11 of group A inspection.

c. Steady-state life test (method 1005 of MIL-STD-883) conditions, or equivalent.

(1) Test condition D or E, using the circuit shown on figure 4, or equivalent.

(2) T_A = 125°C minimum.

(3) Test duration: 1,000 hours, except as permitted by appendix B of MIL-M-38510 and method 1005 of MIL-STD-883.

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4.4.4 Group D inspection. Group D inspection shall be in accordance with table of method 5005 of MIL-STD-883. End-point electrical parameters shall be as specifi in table II herein.

4.4.5 Inspection of packaging. Inspection of packaging shall be as specified in MIL-M-38510.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Voltage and current. All voltages given are referenced to the microcircuit ground terminal. Currents given are conventional current and positive when flowing into the referenced terminal.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be as specified in MIL-M-38510.

6. NOTES

6.1 Notes. The notes specified in MIL-M-38510 are applicable to this specification.

6.2 Intended use. Microcircuits conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.3 Ordering data. The contract or purchase order should specify the following:

a. Complete part number (see 1.2).

b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

c. Requirements for certificate of compliance, if applicable.

d. Requirements for notification of change of product or process to the contracting activity in addition to notification to the qualifying activity, if applicable.

e. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action, and reporting of results, if applicable.

f. Requirements for product assurance options.

g. Requirements for special carriers, lead lengths, or lead forming, if applicable. Unless otherwise specified, these requirements will not apply to direct purchase by or direct shipment to the Government.

h. Requirements for "JAN" marking.

6.4 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510, MIL-STD-1331, and as follows:

F_{MAX}	Maximum operating frequency
GND	Electrical ground (common terminal)
I_{IN}	Current flowing into an input terminal
V_{IC}	Input clamp voltage
V_{IN}	Voltage level at an input terminal

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6.5 Logistic support. Lead materials and finishes (see 3.3) are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class B (see 1.2.2) and lead finish C (see 3.3). Longer length leads and lead forming shall not affect the part number.

6.6 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

Military device type	Generic-industry type
01	54LS390
02	54LS393
03	54LS490

6.7 Manufacturers' designations. Manufacturers' circuits included in this specification are designated as shown in table IV.

TABLE IV. Manufacturers' designations. 1/

Device type	Manufacturer			
	Texas instruments	Signetics	Motorola	Fairchild
	Circuit A	Circuit B	Circuit D	Circuit E
01	X	X		X
02	X	X	X	X
03	X	X		X

1/ An "X" indicates the manufacturer has submitted a schematic for the indicated device type.

6.8 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Air Force - 11, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC, OS, SH
Air Force - 19

Agent:
DLA - ES

Preparing activity:
Air Force - 17

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